DE MORGAN CONVEXITY AND THE STRICT–WEAK COMPARISON IN CUBICAL TYPE THEORY

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Abstract. A companion paper proved that in cubical type theory the *strict fillers* of a boundary — the fillers definable by pure reparametrization, without Kan operations — need not be unique. This paper shows that the non-uniqueness resolves *inside* the strict layer: the De Morgan convex combination

$$\chi(k) \;=\; (\varphi \wedge \neg k) \vee (k \wedge \varphi') \vee (\varphi \wedge \varphi')$$

yields a strict homotopy rel boundary between any two same-boundary fillers of one cell, and, iterated, makes the fiber of strict fillers over a boundary strictly contractible — the exact strict parallel of the uniqueness of Kan fillers up to homotopy. The third term is essential: the naive combination $(\varphi \wedge \neg k) \vee (k \wedge \varphi')$ fails *precisely* by the absence of excluded middle in the free De Morgan algebra, and the failure is machine-checked as a typing error. Fillers originating from different cells admit no direct strict homotopy, but connect by deformation onto the shared subcomplex (a machine-checked zig-zag). The strict–weak comparison then closes a circle opened by the first paper of this series: the constant-tube composition equation, which its no-go theorem shows can never be definitional, holds *propositionally and uniformly* via the canonical filler — what cannot be definitional is exactly what hfill supplies propositionally — while genuinely weak composites open new propositional classes (winding). All positive and negative instances are machine-checked in a self-contained proof-assistant kernel, and every statement is a decidable, assumption-free fact about the algorithmic equality of a minimal fragment.

1. Introduction

This is the fourth paper in a series on the definitional (strict) structure of cubical type theory [1], written, like its predecessors, to be read independently. The first paper [3] developed the metatheoretic platform — in particular a no-go theorem: extending definitional equality with *constant-tube collapse* for homogeneous composition (the equation we call (†) below) is impossible in principle — and the decidability of the algorithmic equality on a minimal fragment. The second [4] characterized the strict layer

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of generic loops as a wedge of De Morgan spheres. The third [5] extended the characterization to cells with generic boundaries — the strict layer of a cellular context realizes its cube presheaf on the nose — and isolated a phenomenon absent from the Kan world: *strict fillers of a common boundary need not be unique.*

The present paper is about what that non-uniqueness means, and more generally about how the strict layer sits inside the full theory. Its results:

(1) *De Morgan interpolation* (Section 3). For interval formulas φ, φ' whose restrictions to every face of the cube agree, the *De Morgan convex combination*

$$\chi(k) := (\varphi \wedge \neg k) \vee (k \wedge \varphi') \vee (\varphi \wedge \varphi')$$

satisfies $\chi(0) = \varphi$, $\chi(1) = \varphi'$ and is constant in k on every face. Consequently any two strict fillers of one boundary arising from the same cell are connected by a *strict* homotopy rel boundary — no Kan operation is involved (machine-checked on the non-uniqueness pair of [5]).

(2) *Necessity of the third term* (Section 3). The naive combination $(\varphi \wedge \neg k) \vee (k \wedge \varphi')$ restricts on a face with common value ρ to $\rho \wedge (k \vee \neg k)$, which is *not* ρ: the free De Morgan algebra has no excluded middle. The would-be homotopy fails to typecheck (machine-checked failure). The correction term $\varphi \wedge \varphi'$ is the exact price of the missing excluded middle.

(3) *Strict contractibility of fiber towers* (Section 4). Iterating the interpolation — two homotopies between the same fillers again share a boundary — the entire tower of strict fillers, homotopies, homotopies between homotopies, ... over a fixed boundary is strictly contractible. Strict fillers are therefore *unique up to strict homotopy*: the exact parallel, inside the strict layer, of the Kan world's uniqueness of fillers up to homotopy.

(4) *Cross-cell connection through the shared subcomplex* (Section 5). Fillers of one boundary originating from *different* cells admit no direct strict homotopy (a strict cube is single-cell material), yet connect by strict deformations onto the shared boundary material — a machine-checked zig-zag $S_1 \rightsquigarrow S_m$, $S_2 \rightsquigarrow S_m$.

(5) *The comparison closed* (Section 6). The constant-tube equation (†), provably never definitional [3], holds propositionally and uniformly: the canonical filler is a path rel boundary from the base to the composite (machine-checked). Conversely the weak layer genuinely extends the propositional classification: the composite $\mathsf{loop} \cdot \mathsf{loop}$ on the circle has winding $+2$, separating it propositionally from the basic strict cells (winding $0, \pm 1$, machine-computed).

The emerging picture is summarized in Table 1: over a common boundary, strict cells are unique up to strict homotopy; constant-tube composites are definitionally new but propositionally strict; genuine composites are new

cells over a common boundary	definitional	propositional
strict (reparametrizations)	unique up to strict homotopy	likewise
constant-tube composites	separated ((†) no-go)	strict, via the filler
genuine composites	separated	new classes (winding)

TABLE 1. The strict–weak comparison over a common boundary.

outright. The interpolation operator equips the strict fibers with a canonical convex structure — we call it *De Morgan convexity* — which supplies the strict layer with its own homotopy theory, definable entirely in the connection algebra.

Honest positioning. Each ingredient is elementary: the interpolation lemma is three absorption computations, and the propositional half of the comparison is the standard filler. The contribution is the assembly and its exactness — that the interpolant is boundary-constant *exactly* when restrictions agree, that the naive interpolant fails *exactly* by no-LEM, that strict homotopy uniqueness holds *despite* definitional non-uniqueness, and that the definitional/propositional boundary traced by the no-go theorem of [3] is exactly the hfill-image. We know of no prior treatment of the strict layer's internal homotopy theory. As throughout the series, all instances are machine-checked and, by the decidability theorem of [3], decided.

2. Preliminaries

We work in the reparametrization-plus-composition fragment of a CCHM-style cubical type theory [1]: path types over a base type, path abstraction/application, and homogeneous composition hcomp; $\mathrm{DM}(\vec{i})$ is the free De Morgan algebra of interval formulas (no excluded middle), with canonical antichain normal forms. Definitional equality $\equiv$ is the algorithmic equality of the kernel of [3]; we import from the companions:

- (K1) endpoint collapse of path application is normal-form exact, and for dependent path types returns the stored boundary cells [5, K1];
- (K2) conversion of neutral spines is head- and argument-exact [4, K2];
- (K3) on the minimal fragment, evaluation and conversion terminate and $\equiv$ is decidable, with no assumptions [3];
- (K4) *realization* [5]: over a cellular context, strict cubes are classified, injectively, by cells and formula tuples; in particular *equal boundaries force equal face restrictions* for same-cell fillers.

The running example. Over the generic-boundary square computad (vertices, edges p, q, r, s, square Q) [5] exhibits two strict fillers of one boundary:

$$F_1 := \langle i\rangle\langle j\rangle\, Q(i \wedge \neg i)(j), \qquad F_2 := \langle i\rangle\langle j\rangle\, Q\big((i \wedge \neg i) \wedge (j \vee \neg j)\big)(j),$$

non-convertible, both inhabiting $\mathsf{PathP}\,(\langle i\rangle\,\mathsf{Path}\,A\,(r(i\wedge\neg i))\,(s(i\wedge\neg i)))\,p\,p$: the two formulas differ in the free algebra yet restrict equally to all four faces.

3. De Morgan interpolation

Lemma 3.1 (Interpolation)**.** *For $\varphi, \varphi' \in \mathrm{DM}(\vec{i})$ let*

$$\chi(k) \;:=\; (\varphi \wedge \neg k) \vee (k \wedge \varphi') \vee (\varphi \wedge \varphi') \;\in\; \mathrm{DM}(\vec{i}, k).$$

Then:

(1) *$\chi(0/k) = \varphi$ and $\chi(1/k) = \varphi'$;*
(2) *for any face f of the $\vec{i}$-cube on which $\varphi|_f = \varphi'|_f = \rho$: $\chi|_f = \rho$, constant in k.*

Proof. (1) $\chi(0) = (\varphi \wedge 1) \vee 0 \vee (\varphi \wedge \varphi') = \varphi \vee (\varphi \wedge \varphi') = \varphi$ by absorption ($\varphi \wedge \varphi' \leq \varphi$); symmetrically at 1. (2) $\chi|_f = (\rho \wedge \neg k) \vee (k \wedge \rho) \vee \rho = \rho$, again by absorption (the first two joinands are $\leq \rho$). □

Theorem 3.2 (Strict homotopy rel boundary)**.** *Let $F = A_c(\vec{\varphi})$ and $F' = A_c(\vec{\varphi}')$ be strict fillers of the same boundary arising from the same cell c. Then*

$$H \;:=\; A_c(\vec{\chi}), \qquad \chi_t := (\varphi_t \wedge \neg k) \vee (k \wedge \varphi'_t) \vee (\varphi_t \wedge \varphi'_t),$$

with k as the outer dimension, is a strict homotopy rel boundary from F to F'.

Proof. By (K4), equality of the boundaries forces $\vec{\varphi}|_f = \vec{\varphi}'|_f$ on every face f. Lemma 3.1(1) gives the k-endpoints; Lemma 3.1(2), applied coordinatewise, makes every face of H constant in k and equal to the given boundary, so H inhabits the path type over the boundary type from F to F'. (Typing uses only the boundary conditions; interior degeneracies of $\vec{\chi}$ are irrelevant.) Machine witness on the running example: `mixInterpD`, with a control probe and with the separation $F_1 \not\equiv F_2$ re-checked — the homotopy is not trivial. □

Lemma 3.3 (Necessity of the third term)**.** *For k-free $\rho \neq 0$:*

$$(\rho \wedge \neg k) \vee (k \wedge \rho) \;=\; \rho \wedge (k \vee \neg k) \;\neq\; \rho,$$

since $\rho \leq k \vee \neg k$ would require every clause of ρ to contain k or $\neg k$. Consequently the naive interpolant $(\varphi \wedge \neg k) \vee (k \wedge \varphi')$ is not *boundary-constant, and the corresponding term fails to typecheck as a homotopy (machine-checked failure on the running example).*

Remark 3.4. Lemma 3.3 is a small but pointed fact: in a Boolean interval the naive convex combination would do, and the third term is invisible. The free De Morgan algebra's missing excluded middle — the very feature that keeps the connection layer faithful [4] — exacts the correction term $\varphi \wedge \varphi'$ as its price. The corrected operation is the "strong" (Kleene-style) convex combination, and we will see it is exactly what the strict layer's homotopy theory runs on.

4. Strict contractibility of fiber towers

Fix a cellular context, a cell c, and a boundary β.

Definition 4.1 (Fiber tower)**.** $\mathrm{Fib}^0(\beta)$ is the set of strict fillers of β from c; for $x, y \in \mathrm{Fib}^n$, $\mathrm{Fib}^{n+1}(x, y)$ is the set of strict homotopies rel boundary between them (strict cubes one dimension up whose k-faces are x, y and whose remaining faces are the constant extensions of the lower boundary data). The tower is *strictly contractible* if at every level any two elements are connected: $\mathrm{Fib}^{n+1}(x, y) \neq \emptyset$ for all x, y.

Theorem 4.2 (Contractibility)**.** *The fiber tower of a cell over any boundary is strictly contractible.*

Proof. Level 1 is Theorem 3.2. For level $n+1$: two elements $H, H' \in \mathrm{Fib}^n$ are strict cubes $A_c(\vec{\chi}), A_c(\vec{\chi}')$ of the same $(m+n)$-cube boundary — their k-faces are the shared lower-level elements and their remaining faces the shared constant extensions — so by (K4) their tuples' restrictions agree on *every* face, and Theorem 3.2 applies verbatim one dimension up. Induction. □

Remark 4.3. Together with the non-uniqueness theorem of [5], the picture is: *strict fillers are not unique, but are unique up to strict homotopy, coherently at all levels* — the precise strict parallel of Kan filling's uniqueness up to homotopy, achieved without any Kan operation. We call the underlying canonical structure — the interpolation operator as a convex combination on fibers — *De Morgan convexity.*

5. Cross-cell fillers and the shared subcomplex

Fillers of one boundary can originate from different cells: over the shared-edge computad of [5] (squares Q_1, Q_2 sharing the edge m), the squares

$$S_1 := \langle i\rangle\langle j\rangle\, Q_1(i \vee \neg i \vee j \vee \neg j)(j), \qquad S_2 := \langle i\rangle\langle j\rangle\, Q_2(i \wedge \neg i \wedge j \wedge \neg j)(j)$$

both inhabit $\mathsf{Path}\,(\mathsf{Path}\,A\,a_{10}\,a_{11})\,m\,m$, together with the edge square $S_m := \langle i\rangle\langle j\rangle\, m(j)$, and are pairwise non-convertible [5].

Proposition 5.1 (No direct homotopy)**.** *There is no strict homotopy with k-faces S_1 and S_2: by the realization theorem a strict cube is a single cell's material, and its two k-faces are that cell's material or boundary material — never the interiors of two different cells.*

Theorem 5.2 (Connection through the shared subcomplex)**.** *Both interiors deform strictly onto the shared material:*

$$\langle k\rangle\langle i\rangle\langle j\rangle\, Q_1\big((i \vee \neg i \vee j \vee \neg j) \vee k\big)(j) \;:\; S_1 \rightsquigarrow S_m,$$
$$\langle k\rangle\langle i\rangle\langle j\rangle\, Q_2\big((i \wedge \neg i \wedge j \wedge \neg j) \wedge \neg k\big)(j) \;:\; S_2 \rightsquigarrow S_m,$$

— at $k = 1$ the first formula becomes 1 and $Q_1(1)(j) \equiv m(j)$ by the attachment equations of [5]*; dually for the second. Hence the strict fiber is connected by the zig-zag $S_1 \rightsquigarrow S_m$, $S_2 \rightsquigarrow S_m$ — entirely inside the strict layer. (Machine witnesses `mixDeformQ1D`, `mixDeformQ2D`.)*

Remark 5.3 (The general pattern)**.** When same-boundary fillers originate from different cells, the realization theorem forces the boundary to lie in the *shared* subcomplex, and the coordinate of each filler transverse to its shared face then has endpoint restrictions on every face — so Lemma 3.1 applies against the corresponding constant, deforming each filler onto shared material, where Theorem 4.2 (one dimension down) connects them. We have verified the two-cell, dimension-2 case by machine; the inductive general statement follows the same descent and we record it at the same level of uniformity as the realization theorem itself [5, Rem. 5.4].

6. The strict–weak comparison

The full theory adds the Kan operations. Two theorems close the comparison over a common boundary.

Theorem 6.1 ((†) is propositional)**.** *Let* $T := \langle j\rangle\,\mathsf{hcomp}\,[\,j{=}0 \mapsto a,\ j{=}1 \mapsto b\,]\,(p\,j)$ *be the constant-tube composite over a generic edge* p *— the left-hand side of the equation* (†) *that the no-go theorem of* [3] *proves can never be definitional. Then the canonical filler*

$$\langle k\rangle\langle j\rangle\,\mathsf{hcomp}\,[\,j{=}0 \mapsto a,\ j{=}1 \mapsto b,\ k{=}0 \mapsto p\,j\,]\,(p\,j)\ :\ \mathsf{Path}\,(\mathsf{Path}\,A\,a\,b)\,p\,T$$

is a path rel boundary from p *to* T*: at* $k = 0$ *the added face is decided and the composite collapses to* $p\,j$*; at* $k = 1$ *the false face is discarded, leaving* T*. The same two computations give the statement for any constant-tube system in any dimension. (Machine witness* `mixFillD`*, together with the standing definitional separation* $T \not\equiv p$*.)*

Remark 6.2. Theorem 6.1 is the exact dual of the no-go theorem: *what can never be definitional is precisely what* hfill *supplies propositionally, uniformly.* The definitional/propositional boundary for composition is not a deficiency to be engineered away — the no-go theorem says it cannot be — but a structural line, with the canonical filler standing exactly on it.

Proposition 6.3 (The weak layer extends the propositional classification)**.** *In the circle instance, the composite* $\mathsf{loop}\cdot\mathsf{loop}$ *has winding number* $+2$*, while* refl*,* loop *and its reversal have winding* $0, +1, -1$ *(all machine-computed in the artifact library of* [3]*). Since winding is a propositional invariant, genuine composites are not propositionally equal to these strict cells: the weak layer adds new propositional classes over the same boundary.*

Table 1 now stands proved in all three rows.

7. Machine verification

All instances are elaboration-time checks in the self-contained Lean 4 cubical kernel of [3]: repository

`https://github.com/r-shibusawa/cubical-strict-j`.

The probes of this paper are the file `LibMixed.lean` — the interpolation homotopy with its control and separation, the machine-checked *failure* of the naive interpolant, the two deformation homotopies of the zig-zag, and the canonical filler with its definitional-separation control — together with the proof notes `docs/DeMorganConvexity.md` and `docs/MixedLayer.md`. They are included from release `v1.4.0`, archived at DOI `10.5281/zenodo.TODOE`. A successful `lake build` replays every check; by (K3) each check is a decision.

8. Related work and outlook

The uniqueness of Kan fillers up to homotopy is foundational for cubical models [1]; the present paper locates its exact strict shadow, running on the connection algebra alone. The series' platform and characterizations are [3, 4, 5]; winding numbers and the circle computation descend from the encode–decode method [2].

Two directions seem natural. First, *a strict model structure in miniature*: De Morgan convexity gives the strict layer contractible fibers and canonical homotopies; whether this organizes into a factorization system or a fibration structure on cellular presheaves — with the strict layer as the "cofibrant skeleton" of the full theory — deserves study. Second, *normal forms up to homotopy*: combining Theorem 6.1 with the realization theorem suggests rewriting genuinely weak cells to canonical composites of strict cells, a step toward word problems for cubical polygraphs [5, §10].

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